NLP Project Proposal - SMS Spam Detection

Course and Section	ARTI 404 / 7MA2	Group Number	9	Instructor	Dr. Mohammed Mosuily

Role	Name	Student ID	Responsibility
Team Leader	Mohammed Nasser Almarhoon	2240005893	Data cleaning and splits; TF-IDF + Logistic Regression; Method 1 analysis; two labelled examples
Member 2	Ali Zuhair Alsaffar	2240005706	Sentence-BERT + Logistic Regression; evaluation; Try it; GitHub/Colab; Method 2 analysis; two labelled examples

1. Task , input and output

Input: one short English SMS message. **Output:** Spam or Not spam, e.g. “You won a cash prize. Pay a fee now to claim it” → Spam. Intended users: mobile users, telecom providers, and small businesses that need to filter unwanted SMS messages quickly. The system will classify unfamiliar messages without retraining.

2. Dataset

We will use the **UCI SMS Spam Collection** (Almeida et al., 2011) [1], containing about 5,574 English SMS messages labelled **ham (0)** or **spam (1)**. The classes are imbalanced (approximately 4,827 ham and 747 spam), so we will use class_weight='balanced' where appropriate and report **Accuracy, Macro F1, and a Confusion Matrix**. We will remove empty rows and exact duplicates before splitting. After cleaning, we will compute the exact number of examples and split the data about **70/15/15** into training, development, and test sets, stratified by label with a fixed random seed.

3. Two Methods

Method 1 (traditional): TF-IDF word features using unigrams and bigrams with Logistic Regression (scikit-learn). **Method 2 (pretrained):** embeddings from the pretrained **all-MiniLM-L6-v2 Sentence-BERT** model [2] using sentence-transformers with frozen weights, followed by Logistic Regression. Both methods use the same classifier. Hyperparameters will be selected using the development set only, and the test set will be used once for the final comparison.

4. Challenge profile and experimental question

Challenge: spam messages containing intentional misspellings or obfuscations, such as **fr33, w1n, cl4im, txt,** and **msg**. Legitimate messages may also contain spam-like words such as “free”.**Question:** Which method detects misspelled or obfuscated spam more reliably: **TF-IDF + Logistic Regression** or **Sentence-BERT + Logistic Regression**?We will flag potential challenge examples using fixed heuristics, including digits inside words, repeated characters, no-vowel tokens of **length 3 or more** (e.g. txt, msg), and selected spam-related keywords. We will first count the resulting subset. If it is too small for meaningful comparison, the challenge definition will be refined and reported clearly. Both methods will be evaluated on the full test set and challenge subset.

5. Evaluation and feasiblity

Both methods will be compared on identical test examples using **Accuracy and Macro F1**, with a **Confusion Matrix** and error analysis of at least three incorrect predictions. If fewer errors occur, all errors will be analysed and difficult correct cases will be discussed.The project will be implemented in a **Google Colab notebook and GitHub repository** with sections for Setup, Data, Method 1, Method 2, Results, Try it, and Sources. The “Try it” cell will classify a new SMS without retraining and handle empty input. The dataset is small enough to run on free Colab resources.

6. Related work

Almeida et al. [1] introduced the SMS Spam Collection for studying SMS spam filtering. Reimers and Gurevych [2]introduced Sentence-BERT, which produces fixed-size sentence embeddings suitable for downstream classification. Our investigation compares these representations to determine whether pretrained sentence embeddings handle misspelled or obfuscated spam better than TF-IDF with bigrams.

7. References

[1] T. A. Almeida, J. M. G. Hidalgo, and A. Yamakami, “Contributions to the study of SMS spam filtering: new collection and results,” *Proc. 11th ACM Symposium on Document Engineering (DocEng)*, 2011, pp. 259–262.**[2]** N. Reimers and I. Gurevych, “Sentence-BERT: Sentence embeddings using Siamese BERT-networks,” *Proc. EMNLP-IJCNLP*, 2019, pp. 3982–3992.